

Measuring Segregation Using Convex Functions

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Outline

Broad motivation for the article

- Folks in the math community are turning to quantitative measures of segregation [DMW23, TBK⁺25, HSS25].
- There's actually a pretty interesting and rich literature in quantitative sociology on this set of topics [Rea, RF02], going back all the way at least to [Sim49].
- MAUP is a thing [Ope84], and we'd like measures that behave in predictable ways as we change analytical scale.
- If you start engaging with these ideas systematically and mathematically, then you come naturally to convexity and Bregman divergences, which are what I am here to sell you today.

Figure 1 is a simple checkerboard pattern. What if we actually oriented the entire piece around the checkerboard:

Preliminaries

Diversity

Segregation

Spatial Scale

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- What properties to have? Pielou criteria [Pie77], various axioms from [RF02], etc.

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[Cho25] [Cho17]

[BMDG05]

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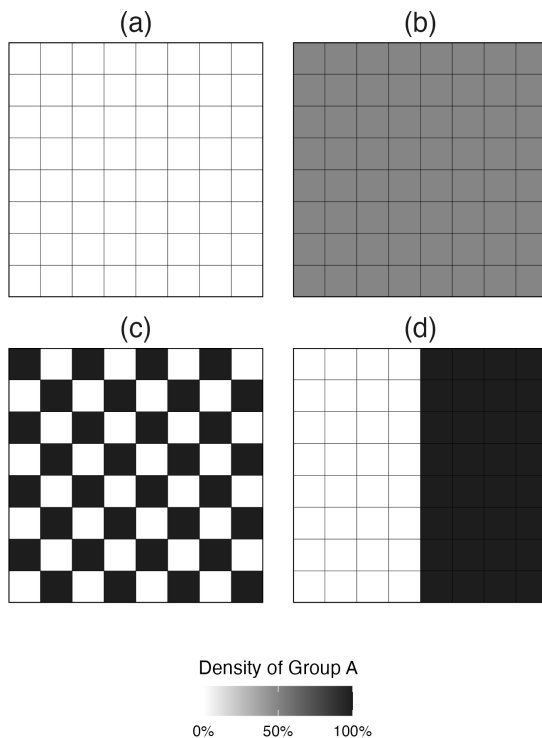


Figure 1: A simple checkerboard pattern.

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TITLE = {The fundamental theorem
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JOURNAL = {Ann. of Math. (2)},
FJOURNAL = {Annals of Mathematics.
Second Series},
VOLUME = {41},
YEAR = {1940},
PAGES = {1--7},
ISSN = {0003-486X},
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MRNUMBER = {0001219},
MRREVIEWER = {R. Brauer},
DOI = {10.2307/1968817},
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